Internet of Things Security

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Internet of Things Security (A35725)

Short Title:	IoT Security		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Forensics and Security		
Author	ADAVY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This course outlines prevailing state-of-the-art practices for IoT security and explains how the IoT security landscape is evolving, to aid IoT stakeholders in making strategic decisions for IoT products, and to help IT professionals strengthen their knowledge about the Internet of Things (IoT) and the security platforms related to it.

Programmes

BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	3 / 6 / M

Indicative Content

- Basic concepts of information security
- IoT System Risks and Challenges
- Challenges for IoT Security
- Basics of Cryptography
- IoT Security Foundation
- Network layer security, IPsec, TLS, 802.11i, Cloud Security, Security in 4G and 5G
- Object sensing layer Security, RFID Security, WSN Security
- IoT Security and Standards

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Understand the security threat landscape and how to take preventative measures.
2. Understand security issues revolving around IoT.
3. Gain knowledge on the security considerations regarding verticals of connected devices.
4. Learn from real-life examples and case studies in IoT Security.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practicals.
- The lectures will be used to introduce new topics and their related concepts.
- The practical element allows the student to put into practice the theoretical concepts covered in the lectures.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	30%	1,2,3
Project	30%	1,2,3
Final Written Examination	70%	1,2,3,4

Assessment Criteria

<40%: Unable to describe key IoT security technologies. Unable to distinguish between different types of IoT device and application vulnerabilities or present instances of them in a clear manner. Unable to effectively use relevant tools.

40%-49%: Can describe in detail key IoT security threats and technologies. Can carry out basic configuration of technologies to implement security policies. Able to present instances of vulnerabilities and carry out threat modelling on a basic IoT system.

50%-59%: In addition to the above, can reason about the various approaches to IoT security and their benefits and limitations. Able to explain in context and present instances of IoT application vulnerabilities. Able to model threats in an IoT system with multiple usage scenarios and actors.

60%-69%: In addition, can explain basis of a variety of cryptographic schemes. Can competently make use of security tools and technologies and carry out effective penetration tests. Able to present and explain how to address IoT application vulnerabilities.

70%-100%: All of the above to an excellent level. Can demonstrate an understanding of some the trade-offs involved in providing security. Able to demonstrate in detail how to address IoT application vulnerabilities.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	24	
Independent Learning	82	

Essential Material

- Li, S. and L. Xu. _Securing the Internet of Things_. China: Syngress, 2017.
- Stallings, W. _Computer Security: Principles and Practice_. 4th ed.. UK: Pearson, 2018.

Requested Resources

- COMPUTER LAB: BYOD Lab